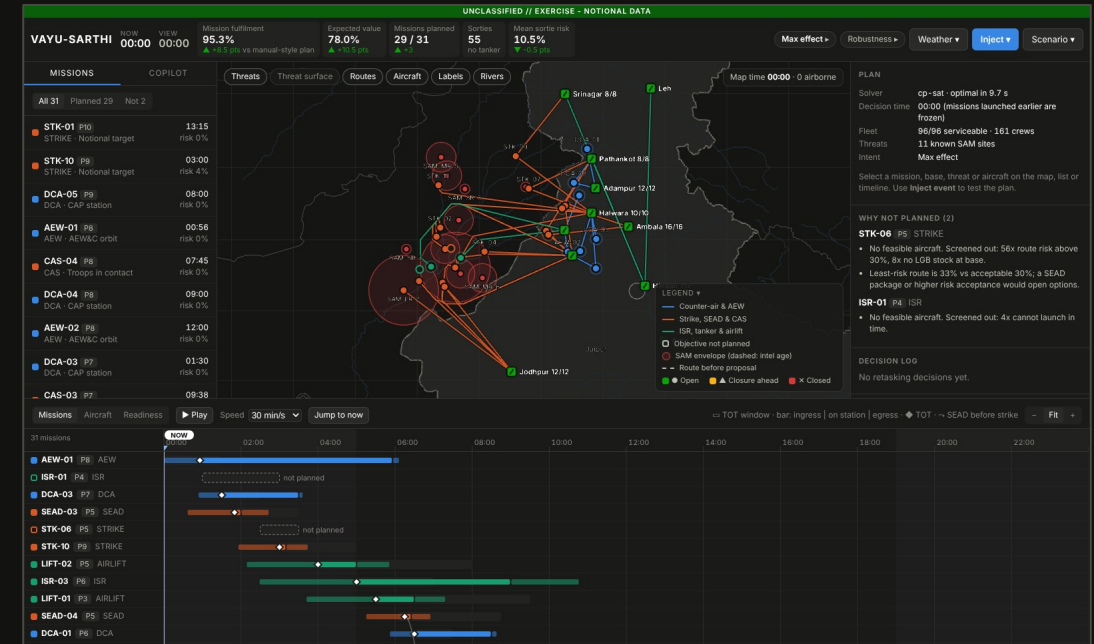


## SMART INDIA HACKATHON 2026 · IDEA PRESENTATION

# VAYU-SARTHI

An explainable decision engine that plans air operations, predicts disruption and re-plans in seconds, changing as little as possible. The commander decides.

|                         |                                                           |
|-------------------------|-----------------------------------------------------------|
| Problem Statement ID    | 26250                                                     |
| Problem Statement title | Air Power: Dynamic Air Operations & Resource Optimisation |
| Organisation            | Ministry of Defence · Defence Services Staff College      |
| Theme                   | Transportation & Logistics                                |
| PS category             | Software                                                  |
| Team ID                 | _____                                                     |
| Team name               | _____                                                     |
| Demo video              | _____                                                     |



Working prototype: common operational picture, synchronisation matrix and KPIs for a 31-mission day.

## MEASURED ON THE PROTOTYPE

**+12.4 pts**

mission fulfillment vs a manual-style plan

**-71%**

aircraft reassigned on a retask vs re-planning from scratch

**70%**

of fog hours forecast (raw weather model: 26%)

## PROPOSED SOLUTION

# Plan, predict, re-plan in seconds, and explain every change

## WHAT IT IS

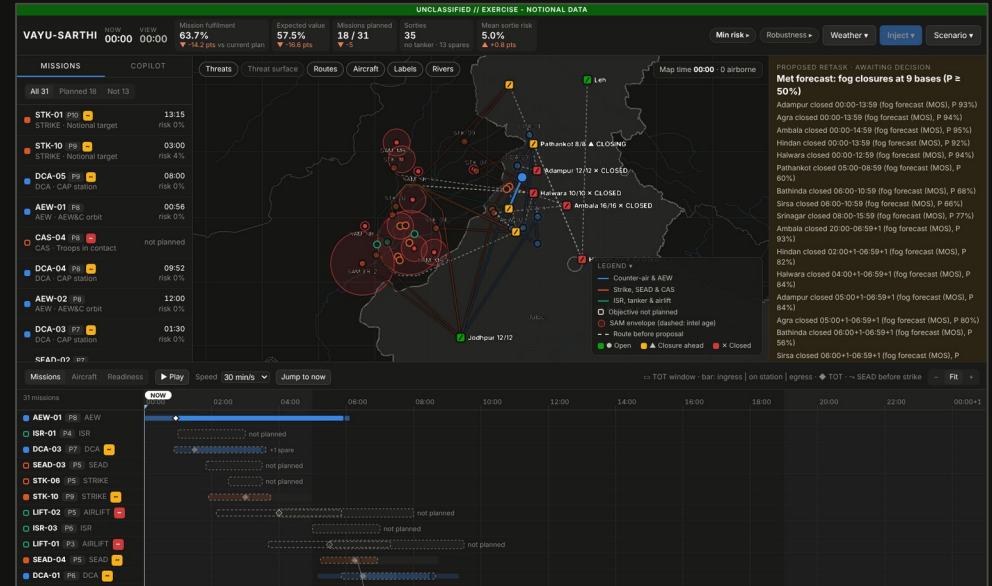
A decision-support system for the air staff. It fuses aircraft, crews, weapons, tankers, weather, threats and airspace into one picture, and computes the best air tasking plan. When something changes it proposes the **smallest set of changes** that recovers the most mission value, as a reviewable diff. A human approves every change.

## HOW IT ADDRESSES THE PROBLEM STATEMENT

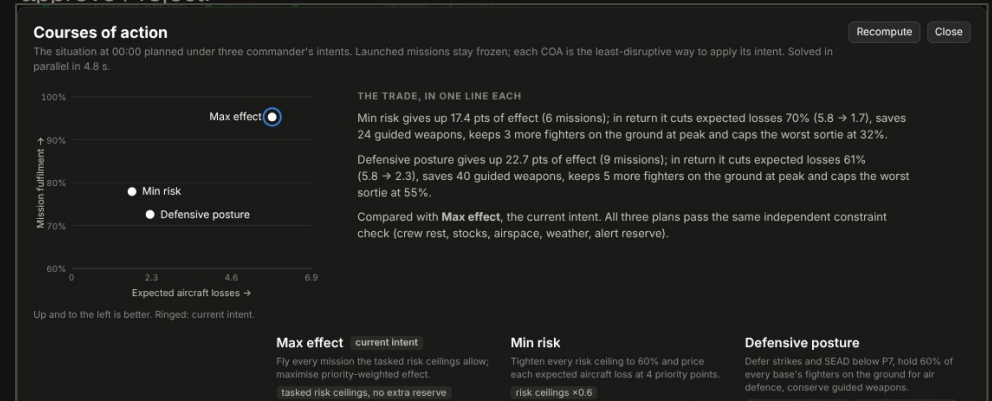
- **Dynamic retasking:** churn-penalised re-optimisation; launched missions frozen; 14.5 vs 49.5 aircraft reassigned for a fogged base, in 2.8 s.
- **Resource optimisation:** one model across aircraft, crew fatigue, weapons, tankers, weather, threats and alert reserve.
- **Weather:** fog forecast trained on real airport observations (Brier skill +44%), turned into probabilistic base closures.
- **Uncertainty:** 2,000 simulated days, single points of failure, ground spares from idle aircraft.
- **Commander's intent:** three courses of action side by side, each trade in one sentence.

## INNOVATION AND UNIQUENESS

- A minimal-disruption diff, not a fresh plan · one integrated optimiser, not seven silos
- Why not, and what it would take, for every unplanned mission · intel-age-aware routing · COAs
- A copilot that answers only from the engine; air-gapped and indigenous; same engine plans flood and earthquake relief



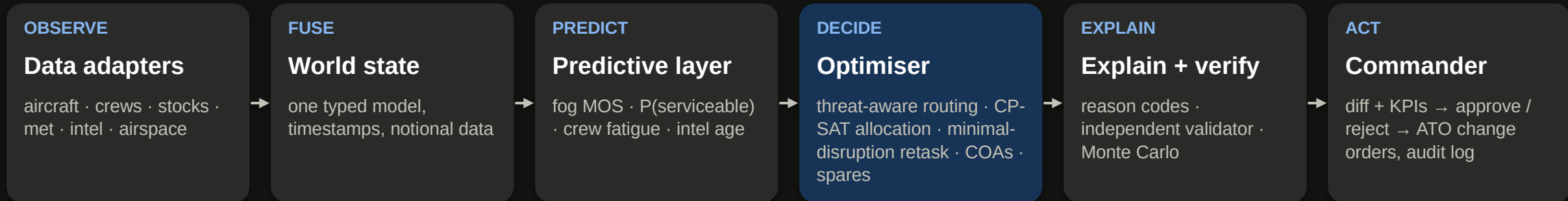
Fog forecast → proposed retask: diff, naive re-plan comparison, KPI deltas, approve / reject.



Courses of action: one situation, three intents, each trade in one line.

## TECHNICAL APPROACH

# Technical approach: an optimiser with a dashboard on top



↻ Events (weather, unserviceability, pop-up threats, new tasking) re-enter the loop; every proposal waits for a human decision.

## TECHNOLOGIES

|                        |                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Decision engine</b> | Python 3.13 · Google OR-Tools CP-SAT · NumPy / SciPy (Dijkstra) · Pydantic                                  |
| <b>Service</b>         | FastAPI (REST); one process serves API and UI; runs offline on one laptop                                   |
| <b>User interface</b>  | React 19 + TypeScript · deck.gl map with an offline basemap (India's official boundary) · Zustand           |
| <b>Prediction</b>      | Logistic MOS on Open-Meteo NWP, labelled with IEM METAR; held-out winter validation                         |
| <b>Copilot</b>         | Parser + optional local LLM (Ollama / llama.cpp) that only picks engine tools; answers are engine templates |
| <b>Assurance</b>       | 55 engine tests · independent constraint validator · Playwright browser end-to-end test                     |

## KEY METHODS

- CP-SAT with optional intervals for aircraft, crews and tankers; cumulative alert-reserve floor; time-on-target domains from crew rest, night currency and base closures
- Retask objective rewards keeping what is planned, weighted by time to launch (×1 planned, ×3 briefed, ×8 armed); launched missions are frozen
- Threat hazard field with intel-age envelope growth and SEAD suppression; Dijkstra plus any-angle smoothing
- Two-process fatigue model gives each crew's fit time windows
- One mission-success model (serviceability with spares, tankers, SEAD before strike, ingress risk) drives both the expected-value KPI and the Monte Carlo

### FEASIBILITY AND VIABILITY

# Feasibility and viability: the prototype already works

### MEASURED RESULTS (NOTIONAL SCENARIOS, LAPTOP CPU)

| Metric                                         | Baseline           | VAYU-SARTHI                    |
|------------------------------------------------|--------------------|--------------------------------|
| Priority-weighted fulfilment · 20 scenarios    | 85.1% manual-style | <b>97.5%</b>                   |
| Expected mission value on the day              | 65.3%              | <b>78.8%</b>                   |
| Aircraft reassigned, base fogged · 8 scenarios | 49.5 naive re-plan | <b>14.5 in 2.8 s</b>           |
| Fog hours forecast · held-out winter           | 26% raw model      | <b>70% (AUC 0.91)</b>          |
| Bad-day (p05) success · ground spares          | 66.9%              | <b>69.9%, 0 flying changes</b> |
| Expected losses · Min-risk COA                 | 5.81 Max effect    | <b>3.01</b>                    |
| HADR relief tonnage · flood / earthquake       | 81.6% / 81.0%      | <b>94.5% / 88.1%</b>           |

Every plan in the tests and benchmarks passes an independent constraint checker (crew rest, stocks, reserve, weather, airspace). Plans in 4–10 s for 35–40 missions; retasks in 1–3 s.

### RISKS AND HOW WE HANDLE THEM

#### Classified data, legacy systems

Adapter per source behind one world model; notional data until access; IACCS/AFNet are adapters, not a rewrite.

#### Trust in machine advice

Nothing changes without approval; a reason for every decision; independent validator; DRDO ETAI principles.

#### Theatre scale

Large-neighbourhood search and decomposition by sector and time; retask touches only the affected missions.

#### Model drift (weather, maintenance)

Probabilities, not yes/no; calibrated and verified; retrained per region and season.

#### Degraded links (DDIL)

Runs air-gapped; an edge node per base syncs on reconnect.

## IMPACT AND BENEFITS

## Impact and benefits

## Operational

- Re-planning goes from hours to seconds, and the plan stays stable: one third of the changes
- More mission value from the same fleet: +12.4 pts fulfilment
- Safer: crew fatigue, night currency and threat risk are constraints, not afterthoughts
- A known bad day: what fails, why, and which aircraft the plan leans on

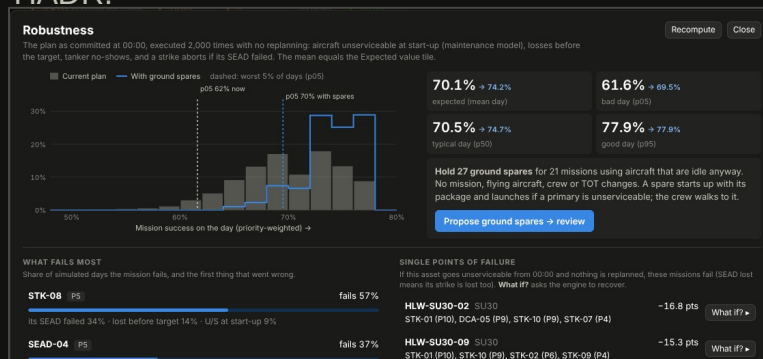
## Resources and cost

- Sorties, tanker sorties and guided weapons are used only where they buy effect
- Commander's intent trades effect for losses and stocks explicitly (Min risk: half the expected losses)
- Idle aircraft become ground spares at no cost to the plan

## National and dual-use

- Same engine for flood relief: 81.6% → 94.5% of relief tonnage with the same fleet; earthquake in the Himalaya 81.0% → 88.1%
- Rescues first; helicopters where there is no runway, derated for thin air; flights stay in Indian airspace
- Indigenous and air-gapped: no foreign cloud or map service; runs on one laptop

**Who uses it:** air tasking order cells at Air HQ and Commands, base operations, air defence controllers, and NDMA / state disaster-response coordination during HADR.

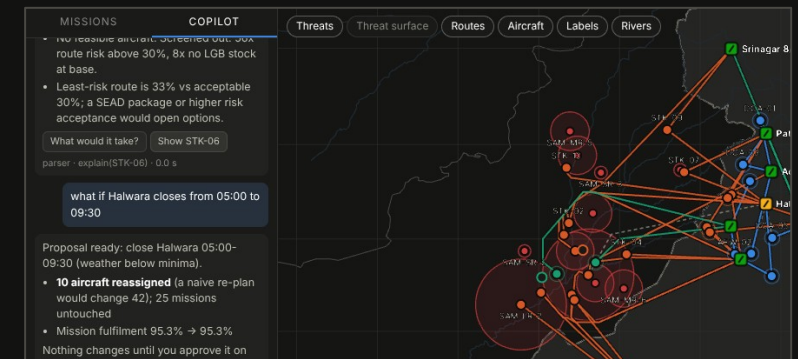


Robustness: 2,000 simulated days, with and without spares

VAYU-SARTHI · SIH 2026 · PS 26250



Flood relief: a breach becomes a P10 rescue



Copilot: answers only from the engine

## RESEARCH AND REFERENCES

# Research and references

**PROBLEM AND CONTEXT**

- IAF AI decision-support tools for IACCS (IDRW, [idrw.org/?p=383033](http://idrw.org/?p=383033))
- DRDO ETAI framework for trustworthy AI in defence (IndiaAI, 2024)
- DARPA Adapting Cross-Domain Kill-Webs ([darpa.mil](http://darpa.mil))
- Kessel Run: KRADOS and Slapshot, ATO in the cloud (Air & Space Forces Magazine)
- Flexible Use of Airspace in India (ICAO APAC IP08, 2023)

**METHODS**

- Perron & Didier, CP-SAT / Google OR-Tools ([developers.google.com/optimization](https://developers.google.com/optimization))
- Glahn & Lowry (1972), Model Output Statistics, J. Applied Meteorology 11
- Borbély (1982), two-process model of sleep regulation, Human Neurobiology 1
- SAFTE fatigue model overview (ICAO FRMS)
- Minimal-deviation rescheduling under disruption (METU, naval task groups)

**DATA (PUBLIC)**

- Open-Meteo historical forecast and forecast APIs ([open-meteo.com](https://open-meteo.com))
- Iowa Environmental Mesonet METAR archive ([mesonet.agron.iastate.edu](https://mesonet.agron.iastate.edu))
- Natural Earth boundaries, India point of view ([naturalearthdata.com](https://naturalearthdata.com))
- Airfield and district headquarters coordinates (public)

**THIS SUBMISSION**

- Working prototype: engine, map, synchronisation matrix, retask review, why-not and what-it-would-take, fog forecast, COAs, robustness, readiness, HADR (flood, earthquake), copilot
- Design and roadmap: [docs/PLAN.md](#) · all scenario data notional